

Muhammad Faizan Khan

Research Interests: Time Series Forecasting | Large Language Models & RAG | Machine Learning / Deep Learning Engineer | AI Engineer | IoT & LoRaWAN | Computer Vision | Signal Processing | Data Scientist

Gender Male | **Nationality** Pakistan

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CAREER OBJECTIVE

PhD researcher at the Faculty of Electronics and Information Technology, Xi'an Jiaotong University, with a Master's degree in Computer Science & Artificial Intelligence from Xidian University. Current research focuses on deep learning for time series forecasting and on Large Language Models, retrieval-augmented generation (RAG/GraphRAG) and LLM-based knowledge systems. Experienced in developing machine learning, signal processing and connected sensing solutions, including a ChirpStack v4 / MQTT LoRaWAN monitoring prototype with simulated OTAA activation, SQLite persistence and REST-based device health monitoring. Strong publication record as first author and co-author in AI, wireless communications and intelligent systems. Seeking opportunities in AI Engineering, Applied Research, Data Science and Intelligent Systems Development.

EDUCATION

2025 – Present	Xi'an Jiaotong University, Xi'an, China	PhD, Faculty of Electronics and Information Technology	Ongoing
2025	Xidian University, Xi'an, Shaanxi, China	Master in Computer Science & Artificial Intelligence	GPA: 88%

CURRENT DOCTORAL RESEARCH

2025 – Present	Faculty of Electronics and Information Technology, Xi'an Jiaotong University Doctoral research on deep learning for time series forecasting and on Large Language Model based knowledge systems.
Time Series Forecasting	● Research on multi-horizon wind power forecasting for single turbines, wind farms and regional aggregations, using SCADA and public benchmark datasets. ● Reproduced and benchmarked state-of-the-art Transformer based forecasting models (iTransformer, TimeXer, PatchTST, TimesNet, DLinear) on standard benchmarks (ETTM1, ECL, Weather, Traffic) and on real wind farm data, with rigorous persistence and statistical baselines. ● Designed a physics-guided Transformer architecture and a regime-aware ensemble that conditions on wind speed bins, direction sectors and ramp states, improving longer-horizon accuracy over strong baselines. ● Built a reproducible forecasting pipeline covering data cleaning, physical feature engineering, sliding-window construction, leakage-free splits and bootstrap confidence intervals.
LLM, RAG & LLM Wiki	● Working on Large Language Model applications: retrieval-augmented generation (RAG), GraphRAG and multimodal retrieval over private

and domain-specific corpora.

- Developing an "LLM Wiki" knowledge system that compiles a maintained knowledge base from documents, with typed supersession relations, validity intervals on edges, cross-document conflict resolution and provenance binding so that outdated assertions are correctly retired.
- Implemented a working reference prototype with an automated test suite, and evaluated entity extraction, graph construction and Personalized PageRank based retrieval.
- Exploring LLMs for time series forecasting, combining language model reasoning with numerical forecasting backbones.

Group Work

Present papers at bilingual (Chinese/English) group reading meetings, prepare technical reports and reproduction studies, and contribute to a client-facing wind power forecasting project with regular progress reviews.

RESEARCH WORK & PUBLICATIONS

Sept 2023 – July 2025

The State Key Laboratory of Artificial Intelligence, Xidian University

Authored six research papers in Machine Learning and Artificial Intelligence: first author on three papers and co-author on the remaining ones. Work focused on designing and improving deep learning architectures, including a novel CNN-GRU model, to enhance classification performance and model robustness under complex and noisy conditions.

First Author

- *Hybrid Modulation Classification: Traditional Signal Processing and LSTM* — ICMMT 2025
- *Automatic Modulation Classification using a Deep Learning Model based on ResNet*
- *A Convolutional Neural Network and Particle Swarm Optimization Hybrid Model for Scripting Language Handwritten Character Recognition*

Co-Author

- *Green HRM and STARA Capabilities in Higher Education: The Mediating Roles of Green Innovation and Green Passion in Sustainable Performance* — International Journal of Sustainability in Higher Education, Emerald, 24 August 2026, pp. 1–35. With M. W. Nazir, Z. Shen and L. A. Rahoo. <https://doi.org/10.1108/IJSHE-01-2026-0113>
- *Accurate Detection of Heart Disease Using the CardioFusion Ensemble Algorithm* — ICECE 2024
- *Generating Synthetic Chest X-ray Images for Improved COVID-19 Detection Using Generative Adversarial Networks* — UCOM 2024

Thesis

Master's Thesis: *Enhancing Automatic Modulation Classification Using Multi-Stream Deep Learning Architectures*. Two conference papers published and one accepted for publication in an IEEE conference based on this research.

Seminars & Courses

Attended academic and technical seminars including Unicom-2023, Unicom-2024 and AI-6G Communication. Coursework in Machine Learning, Deep Learning, Algorithm Analysis and Design, and Data-Driven Optimization. Online courses: Harvard CS50's Artificial Intelligence with Python, Machine Learning and Deep Learning, MATLAB (official platform) and Python (freeCodeCamp).

PROFESSIONAL EXPERIENCE

Research Assistant & Academic English Writing Specialist

Freelance support in academic writing, research editing and technical proofreading for students, scholars and researchers across biomedical sciences, chemistry, AI and technology. Structure scientific manuscripts, improve grammar and clarity, and align texts with journal submission standards (IEEE, Elsevier). Collaborate on research documentation, project reports, literature reviews and grant proposals.

Freelance Web Developer — Fiverr & Upwork (2017 – 2021)

- Delivered more than 50 projects for clients across Europe, North America, the Middle East and Asia, maintaining strong ratings and repeat business over four years on both platforms.
- Designed, built and deployed the institutional website for Government College of Technology (GCT), Faisalabad, covering information architecture, responsive front-end, content management and hosting setup.
- Work spanned custom websites and landing pages, WordPress themes and plugin configuration, PHP/MySQL backends, e-commerce and CMS setup, site migration, performance optimisation, SEO basics and ongoing maintenance.
- Handled the full client cycle independently: requirement gathering, quoting, delivery to deadline, revisions and post-launch support, working with international clients across time zones in English.

Web & Backend Development

- Build REST APIs and AI model serving endpoints with Python and FastAPI, including request validation with Pydantic, asynchronous endpoints, background tasks and automatic OpenAPI/Swagger documentation.
- Deploy machine learning and LLM inference services behind FastAPI, containerised with Docker and served with Uvicorn/Gunicorn; integrate vector databases and retrieval pipelines for RAG applications.
- Full-stack experience with Node.js, PHP, WordPress and SQL/NoSQL databases (MongoDB, MySQL); familiar with Git-based workflows and Linux server environments.

IoT & LoRaWAN Engineering Project (2026)

- Built a Dockerized IoT monitoring system integrated with ChirpStack v4 and MQTT, with SQLite persistence, REST endpoints, online/stale device tracking, historical telemetry charts and health diagnostics.
- Validated an end-to-end simulated LoRaWAN workflow from OTAA activation and gateway traffic through ChirpStack application uplinks to the monitoring dashboard; added automated tests for decoding, ingestion, persistence, staleness and message handling.

Techlogix Pakistan

Completed an 8-week internship with the Banking Implementation Team on the Flexcube project at Askari Bank. Worked on backend development in Node.js and C#, Flexcube integration in Java, and testing of banking modules. Gained exposure to MVVM architecture, WPF/WCF applications, real-time forex systems, MT4/DDF integration and SQL database management.

NADRA Pakistan

Completed a 6-week internship with the Technical Support Team, gaining hands-on experience in NADRA's proprietary NTS system, database management with MongoDB and application development using Node.js. Developed troubleshooting skills for OS and network related issues.

Technical Support – IT Solutions

Provide technical support for web applications, hosting environments and general IT systems. Skilled in diagnosing software, hardware, server and network issues; remote assistance for WordPress, PHP-based systems, CMS platforms, control panels (cPanel, Plesk), email configuration and database management.

TECHNICAL SKILLS

Programming	Python (primary), MATLAB, Java, C#, Node.js/JavaScript, PHP, SQL
AI / ML	PyTorch, TensorFlow, Keras, Scikit-learn, LightGBM/XGBoost, OpenCV, Hugging Face Transformers, LangChain, vector databases (FAISS / Chroma)
Research Areas	Time series forecasting, Large Language Models, RAG / GraphRAG, multimodal retrieval, IoT / LoRaWAN sensing, computer vision, natural language processing, signal processing, automatic modulation classification, federated learning, neural architecture search
Web & Systems	FastAPI, REST API design, Uvicorn, Docker / Docker Compose, MQTT, ChirpStack v4, SQLite, Git/GitHub, Linux, cloud and GPU compute platforms (AutoDL, RTX 4080 SUPER workstation)
Tools	LaTeX/Overleaf, draw.io, Multisim, Proteus, Microsoft Word, Excel and PowerPoint

CERTIFICATIONS

- Machine Learning – Stanford University (Coursera)
- Neural Networks and Deep Learning – DeepLearning.AI (Coursera)
- CS50's Introduction to Artificial Intelligence with Python – Harvard University

PERSONAL SKILLS & ACTIVITIES

Communication	Proficient in articulating complex research concepts, methodologies and results. Experienced in authoring research papers and publishing in peer-reviewed venues, preparing comprehensive technical reports, and presenting bilingual (English/Chinese) technical talks at group and conference level.
Leadership & Volunteering	● Punjab Youth Festival Ambassador, Faisalabad Region ● IEEE University Chapter, Volunteer ● Vice Manager, University Tech Fair ● Volunteer, International Student Organization, Xidian University — organised cultural events and workshops fostering cross-cultural connections ● Football team player, International School — teamwork and strategic play

ACHIEVEMENTS

- Awarded fully-funded Chinese Government Scholarship (CSC) for Master's studies

LANGUAGES

Fluent in English, Urdu, Pashto and Hindi.
Chinese — intermediate level, HSK-3: 264/300.

REFEREES / MENTORS

Prof. Shuyuan Yang — International Research Center of Intelligent Perception and Computation, School of Artificial Intelligence, Xidian University, Xi'an, China · syyang@xidian.edu.cn

Prof. Zhixi Feng — International Research Center of Intelligent Perception and Computation, School of Artificial Intelligence, Xidian University, Xi'an, China · zxfeng@xidian.edu.cn